



PLATFORM 9

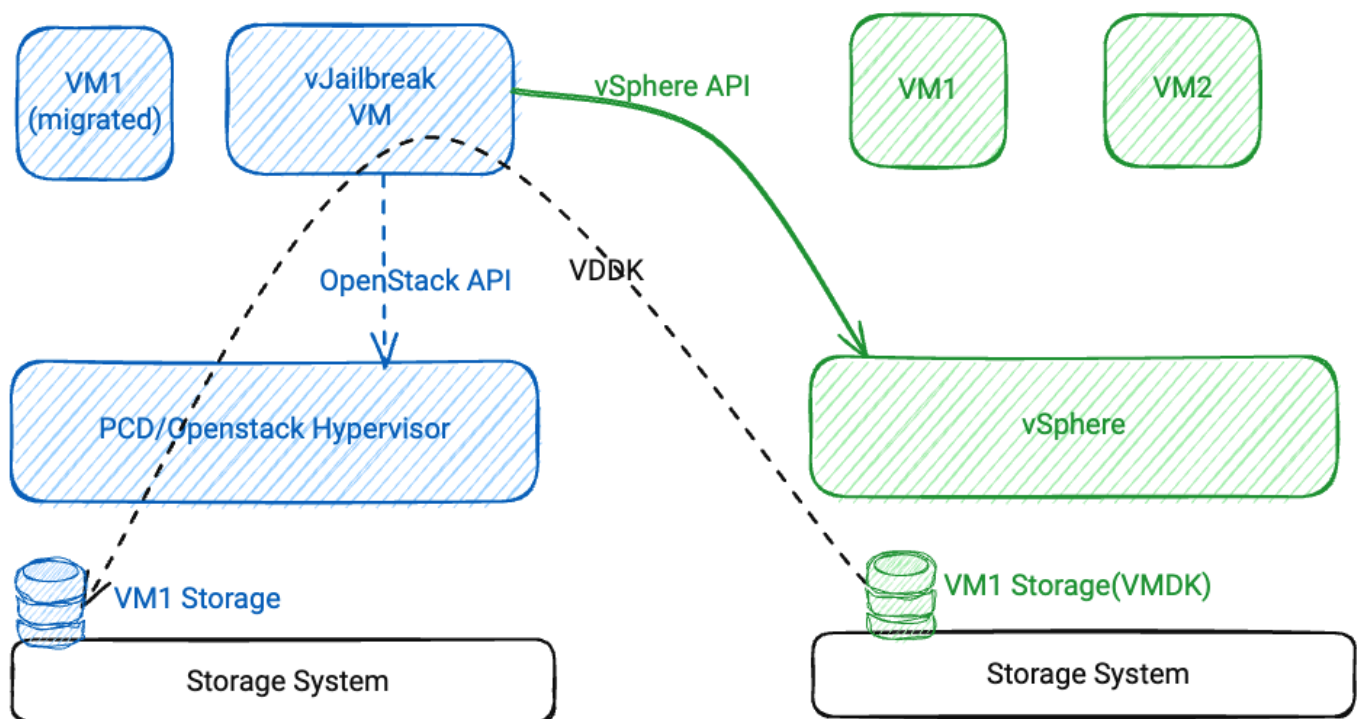
Prepared by: S Arvind (Solutions Architect) Version: v1.1_202601

Objective

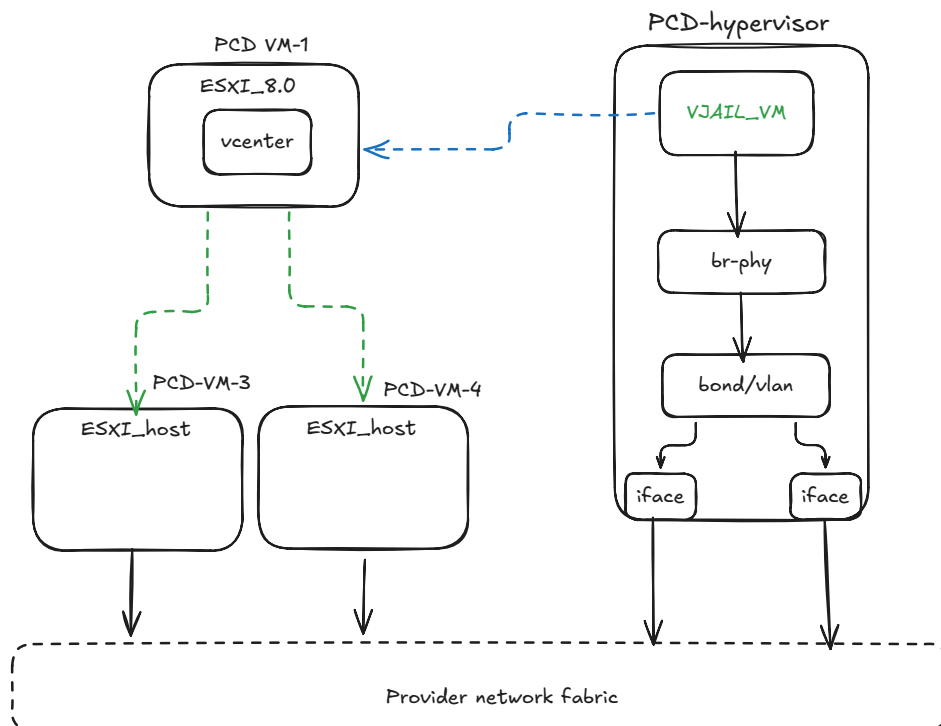
- Outline of Configurations and settings for setting up vjailbreak environment

General architecture layouts:

Overview



Network architecture



Deployment and configuration

Pre-requisites

Environment

- PCD environment should be ready and deployed with required backing store and networks.
- vjailbreak image can be deployed as ephemeral or block store backed VM instance in pcd (preferably block store volume)
- The VM should be deployed in the network which has access to VMware and the pcd environment.

Access and credentials:

- VMware credentials used for by the tool would need to have the policies applied as per doc:
<https://platform9.github.io/vjailbreak/introduction/prerequisites/>
- For PCD credential an appropriate rc file to be supplied which has pcd local user with admin role for the tenants.

Installation

- On the pcd environment download the latest vjailbreak image on the jump host that can access the pcd via command line, by following the guide:
https://platform9.github.io/vjailbreak/introduction/getting_started/
- Run the openstack command to upload the image into the glance library:

```
openstack image create --os-interface admin --insecure --container-format bare --disk-format qcow2 --file vjailbreak-image.qcow2 vjailbreak-0.3.8
```

- Deploy an instance in pcd connected to the required provider network. The VM should have a flavor which has **minimum 16 core and 32GB RAM with 150GB** boot storage.
- During VM deployment from the PCD UI, select the *cloud-init* flag and leave the values as **default** in the associated field.

Configuration

- Access the VM over ssh
- Copy the latest version of the for Linux into `/home/ubuntu` of the vJailbreak VM. Untar it to a folder named `vmware-vix-disklib-distrib` in the `/home/ubuntu` directory.

```
# Example: Extract VDDK tarball
cd /home/ubuntu
tar -xvf VMware-vix-disklib-8.0.3-*.x86_64.tar.gz
# This will create the vmware-vix-disklib-distrib directory
```

- Set the `/etc/host` you can add them manually later.

```
# Example manual addition to /etc/hosts
sudo sh -c 'echo "192.168.1.100 vcenter.example.com" >> /etc/hosts'
sudo sh -c 'echo "192.168.2.100 openstack.example.com" >> /etc/hosts'

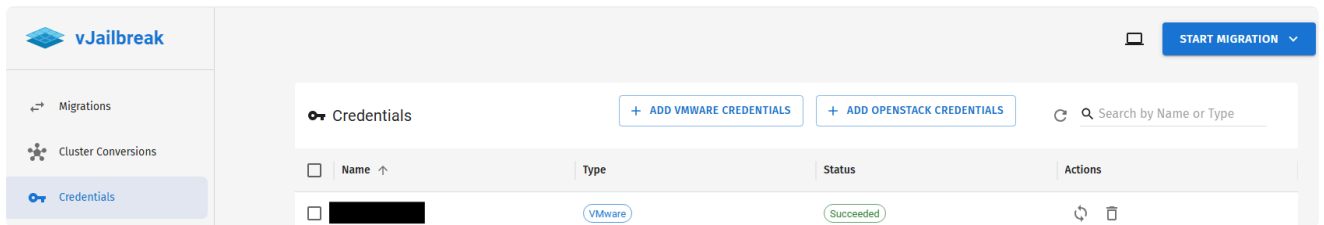
# ESXi hosts entries (required for VM copy phase)
sudo sh -c 'echo "192.168.1.101 esxi01.example.com esxi01" >> /etc/hosts'
sudo sh -c 'echo "192.168.1.102 esxi02.example.com esxi02" >> /etc/hosts'
```

Setting up credential

PCD RC FILE

- Prepare the rc file while the `OS_PROJECT_NAME` pointing to respective tenant for a given region.
- For every tenant there would be one rc file uploaded with the same user having admin rbac for that tenant

- Under the credential section add the VMware and PCD credential:



- For PCD credential toggle the `pcd` credential before upload the rc file:

Add OpenStack Credentials

☐ Is PCD credential

☐ Allow insecure TLS (skip SSL verification)

Environment data prepare for migrate operation

Collecting inventory of VMs from the existing Vmware environment.

- preparing a curated list of VMs containing following fields for every VM that are to be migrated.
 - ip address
 - VM name
 - Specific OS type
 - source Backend storage name
 - Source network name
 - Destination PCD target cluster name
 - **Destination pcd backend store name** (This storage backend should be served by atleast one host in the target cluster)
 - **Destination pcd network/provider** (The provider network should belong to the host config label that the target host cluster nodes are mapped to)
 - comment field
- Identifying and exclusion of any very old legacy VMs running with very old kernel.

- Before the inventory list is populated the targeted VMs should be checked for basic connectivity/reachability etc

Flavor selection

- `Vjailbreak` has the intelligence to select the closest flavor available in PCD that matches the VM resource that are migrated.
- As a best practice, **manually pre-create the required flavors that would generally match VMware instances**. This process should be done all before migration.

Flavor creation

- Any flavor that is created in PCD which will then be scoped by `vjailbreak` tool should have 0 Disk size defined.

Migrate operation workflow

- Review the list of inventory to be migrated to PCD.
- Run the pre-check steps to ensure the VMs are reachable before migration.
- From the UI trigger the migration process according to the list
- Post successful migration test base connectivity and release for user validate.